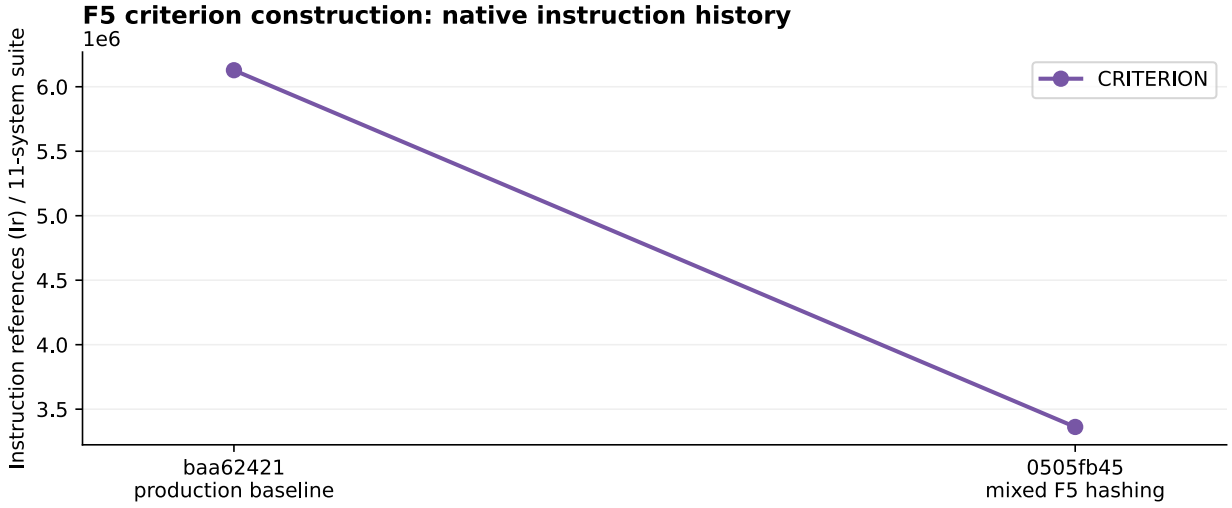
F5 criterion construction

Native instruction and allocation measurements · 2026-10-12



5 paired A/B/A′ rounds; native medians and all samples. Merged worker coverage.
Instrumented complete calls; no wall-time or RSS claim. Source: records/comparison.json.

CRITERION: 6,128,134 → 3,362,140 Ir per suite. Paired instruction reduction 45.14% (95% bootstrap interval 45.121% to 45.155%). Native verdict: fewer instructions.

The baseline profile records standard hashing as 46.68% of criterion self instructions. The candidate uses the existing mixed-mask hasher for monomials and the integer hasher for temporary pivot indices. Its pruning rules, signature order and counted XOR work are unchanged. Standalone criterion cost is a phase result; it is already included in complete F5 calls and is not added a second time.

Native sources: comparison and raw-record hashes, frozen protocol, result report and validation. Raw Callgrind dumps and Rust JSONL remain beside the source data. This 11-system suite is separate from historical n=17/n=23 CI diagnostics. Validation: 3,906 release-library tests passed, 110 ignored, zero failures. The previously unfinished test passed unmodified. All 18 focused F5 tests and five comparison tests passed. Runtime and process RSS remain unmeasured.